

The Untold Story of UNIX

Curious about how and why UNIX was developed?
You'll find the answers here.

When we think about modern computing — multitasking, file systems, portability, and open collaboration — one operating system quietly underpins it all: UNIX. Its legacy shapes Linux, macOS, Android, and even Windows subsystems today. However, few know that this revolutionary system was born not from grand corporate planning, but from a programmer's quest to make a simple space game run smoothly.

Let's take a journey back to where it all began.

The origins: From Multics to a dream

In the mid-1960s, Bell Telephone Laboratories, the General Electric Company (GE), and Project MAC of MIT (Massachusetts Institute of Technology) joined forces to create a groundbreaking new operating system called Multics (Multiplexed Information and Computing Service).

The vision for Multics was ambitious — it aimed to allow simultaneous computer access to a large community of users, offering abundant computing power, massive data storage, and effortless data sharing.

However, when primitive versions of Multics began running on the GE-645 computer in 1969, they failed to deliver the general-purpose computing service that was promised. The system was too



complex, too slow, and development dragged on.

Eventually, Bell Labs decided to withdraw from the project. Their researchers were left without an effective interactive computing environment but with plenty of ideas and determination to build something better.

Ken Thompson and Dennis Ritchie: A new beginning

To improve their programming environment, Ken Thompson, Dennis

Ritchie, and their colleagues at Bell Labs began sketching a paper design for a new file system — an idea that would later evolve into the UNIX file system.

Thompson then wrote programs to simulate how this file system would behave in a demand-paging environment and even encoded a simple kernel for the GE-645 computer to test the concept.

The space travel spark

During this period, Thompson wrote a game called 'Space Travel' in Fortran for the GECOS operating system. Unfortunately, the game was difficult to control, slow, and costly to run.

Then, Thompson discovered an old PDP-7 computer lying around in Bell Labs. It was a modest machine but offered good graphics and cheap execution time. It became the perfect playground for experimentation.

Programming on the PDP-7 gave Thompson a deep understanding of the hardware but there was a major obstacle — it had no development tools. There were no compilers, editors, or assemblers.

To overcome this, Thompson and others wrote and assembled programs on a more powerful GECOS mainframe, a process called cross-assembly, where code is developed on one computer (the host) but intended to run on another (the target).